WEEK 3

3	3	10	Perform the following DB operations using Cassandra. a) Create a keyspace by name Library b) Create a column family by name Library-Info with attributes Stud_Id Primary Key, Counter_value of type Counter, Stud_Name, Book-Name, Book-Id, Date_of_issue c) Insert the values into the table in batch d) Display the details of the table created and increase the value of the counter e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times. f) Export the created column to a csv file g) Import a given csv dataset from local file system into Cassandra column family

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bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {
   ...   'class': 'SimpleStrategy',
   ...   'replication_factor': 1
   ... };
AlreadyExists: Keyspace 'library' already exists
cqlsh> describe keyspaces;

employee    library   students            system_schema
employee1   library1  system              system_traces
employee11  stud      system_auth         system_views
lakshya     student   system_distributed  system_virtual_schema

cqlsh> CREATE KEYSPACE p_library WITH replication = {   'class': 'SimpleStrategy',   'replication_factor': 1 };
cqlsh> USE p_library;
cqlsh:p_library> CREATE TABLE Library_Info (
             ...     Stud_Id int,
             ...     Book_Id int,
             ...     Stud_Name text,
             ...     Book_Name text,
             ...     Date_of_issue date,
             ...     Counter_value counter,
             ...     PRIMARY KEY (Stud_Id, Book_Id)
             ... );
InvalidRequest: Error from server: code=2200 [Invalid query] message="Cannot mix counter and non counter columns in the same table"
cqlsh:p_library>
cqlsh:p_library> DROP TABLE IF EXISTS Library_Info;
cqlsh:p_library> CREATE TABLE Library_Info (
             ...     Stud_Id int,
             ...     Book_Id int,
             ...     Stud_Name text,
             ...     Book_Name text,
             ...     Date_of_issue date,
             ...     PRIMARY KEY (Stud_Id, Book_Id)
             ... );
cqlsh:p_library> CREATE TABLE Library_Counter (
             ...     Stud_Id int,
             ...     Book_Id int,
             ...     Borrow_Count counter,
             ...     PRIMARY KEY (Stud_Id, Book_Id)
             ... );
cqlsh:p_library> BEGIN BATCH
             ... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
             ... VALUES (112, 1, 'Anu', 'BDA', '2026-02-24');
             ...
             ... INSERT INTO Library_Info (Stud_Id, Book_Id, Stud_Name, Book_Name, Date_of_issue)
             ... VALUES (113, 2, 'Rahul', 'DBMS', '2026-02-24');
             ... APPLY BATCH;
```

```
cqlsh:p_library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |       2 |      DBMS |    2026-02-24 |     Rahul
     112 |       1 |       BDA |    2026-02-24 |       Anu

(2 rows)
cqlsh:p_library> UPDATE Library_Counter
             ... SET Borrow_Count = Borrow_Count + 1
             ... WHERE Stud_Id = 112 AND Book_Id = 1;
cqlsh:p_library> SELECT * FROM Library_Counter;

 stud_id | book_id | borrow_count
---------+---------+--------------
     112 |       1 |            1

(1 rows)
cqlsh:p_library> UPDATE Library_Counter SET Borrow_Count = Borrow_Count + 1 WHERE Stud_Id = 112 AND Book_Id = 1;
cqlsh:p_library> SELECT * FROM Library_Counter;

 stud_id | book_id | borrow_count
---------+---------+--------------
     112 |       1 |            2

(1 rows)
cqlsh:p_library> SELECT Borrow_Count
             ... FROM Library_Counter
             ... WHERE Stud_Id = 112 AND Book_Id = 1;

 borrow_count
--------------
            2
```

```
cqlsh:p_library> exit
bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ cqlsh -e "COPY Library.Library_Info TO 'library_info.csv' WITH HEADER=TRUE;"
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 1 rows; Rate:      13 rows/s; Avg. rate:      13 rows/s
1 rows exported to 1 files in 0.087 seconds.
bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ cqlsh -e "COPY Library.Library_Counter TO 'library_counter.csv' WITH HEADER=TRUE;"
Using 16 child processes

Starting copy of library.library_counter with columns [stud_id, book_name, counter_value].
Processed: 1 rows; Rate:      17 rows/s; Avg. rate:      17 rows/s
1 rows exported to 1 files in 0.067 seconds.
bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ ls
 40x              Desktop           filelink.txt                              Library_Info.csv                 PrintNumbers.scala  SimpleTextStreamingCleanerApp  test.txt
 a.py             Documents         file.sh                                   main                             Public              snap                          '"test.txt".odt'
 ayush.sh         Downloads         file.txt                                  Music                            queue.c             spark-3.0.3-bin-hadoop2.7.tgz  Videos
 basics           eclipse-workspace file.txt.save                             pattern.sh                       queue.h             Sparkdata.txt                 WordCountFilter.scala
 b.py             emp.txt           import_library_library_info.err.20250408_150552  pattern.sh.save                  result.txt          student
 c.py             f1                input.txt                                 Pictures                         sample.txt.save     student.dat
 deb              f2                library_counter.csv                       'PrintNumbers$$anonfun$main$1.class'  script              student.txt
 depta.txt.save   file1.txt         Library_Counter.csv                       'PrintNumbers$.class'            shellscript.sh      studnt.dat
 depta.txt.save.1 file2.txt         library_info.csv                          PrintNumbers.class               Simple.scala        Templates
bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ cqlsh -e "COPY Library.Library_Info FROM 'library_info.csv' WITH HEADER=TRUE;"
Using 16 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 1 rows; Rate:       2 rows/s; Avg. rate:       3 rows/s
1 rows imported from 1 files in 0.372 seconds (0 skipped).
bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ cqlsh -e "COPY Library.Library_Counter FROM 'library_counter.csv' WITH HEADER=TRUE;"
Using 16 child processes

Starting copy of library.library_counter with columns [stud_id, book_name, counter_value].
Processed: 1 rows; Rate:       2 rows/s; Avg. rate:       3 rows/s
1 rows imported from 1 files in 0.364 seconds (0 skipped).
bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$
```